

FORM TP 2011045



TEST CODE **01207020**

MAY/JUNE 2011

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

BIOLOGY

Paper 02 – General Proficiency

2½ hours

READ THE FOLLOWING DIRECTIONS CAREFULLY.

- 1. This paper consists of SIX questions in two sections. Answer ALL questions.**
- 2. For Section A, write your answers in the spaces provided in this booklet.**
- 3. For Section B, write your answers in the spaces provided at the end of each question, in this booklet.**
- 4. Where appropriate, answers should be illustrated by diagrams.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1. A student investigating transpiration carried out the following procedure. He obtained two celery stalks and cut the bottom end of each while they were submerged in water. The cut ends of both stalks were then quickly placed into two flasks of dye, Flask A and Flask B. Flask A was placed on a laboratory bench in normal room conditions, while flask B was put in the same room but warm air from a blow-dryer was directed onto it. The time was noted and the celery stalks were allowed to stand for 30 minutes. The student measured and recorded the distance the dye had travelled up each stalk at five-minute intervals during the 30-minute period. The results are shown in Table 1.

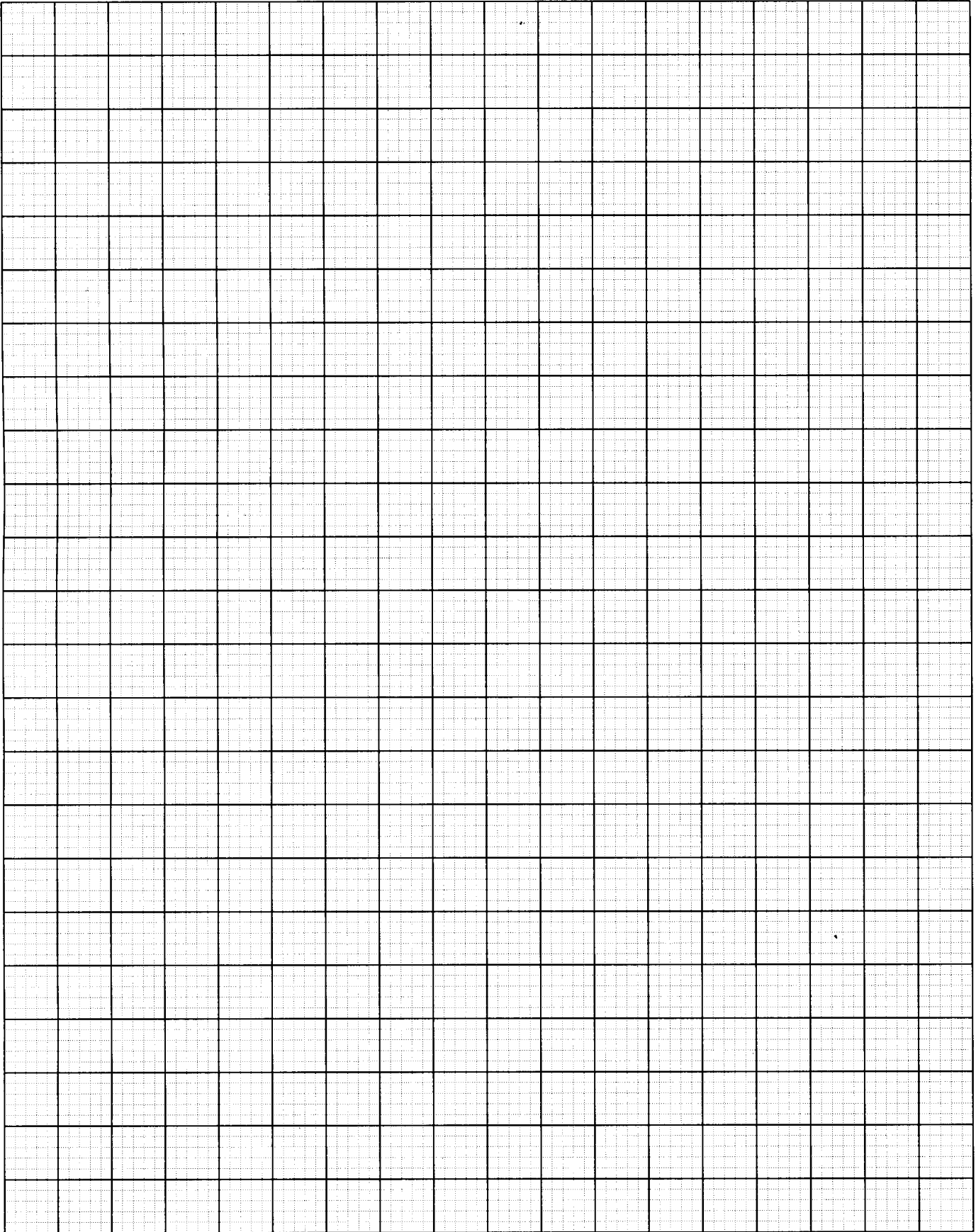
TABLE 1: DISTANCE DYE TRAVELLED UP THE CELERY STALKS

Time (minutes)	Stalk A (distance in cm)	Stalk B (distance in cm)
5	1	2
10	3	4
15	5	8
20	7	12
25	9	15
30	10	17

- (a) (i) **On the grid provided on page 3, draw a line graph to represent the data shown in Table 1. (4 marks)**
- (ii) Account for the differences in the shape of the line graphs for Stalks A and B.

(2 marks)

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- (b) Write a suitable aim for this experiment.

(2 marks)

- (c) Which flask represents the control in this investigation?

(1 mark)

- (d) What conclusion can be drawn from this data?

(2 marks)

- (e) Describe the physical processes involved in the movement of water from the soil into the plant roots, up the stem and out through the leaves.

(4 marks)

- (f) Describe TWO ways in which the xylem is adapted for transporting water within a flowering plant.

(2 marks)

- (g) Suggest THREE ways in which transpiration benefits plants.

(3 marks)

- (h) Name a type of transport tissue, OTHER THAN xylem, found in flowering plants and state the composition of the fluid transported in this tissue.

(2 marks)

- (i) Outline ONE difference and TWO similarities between the transport system of flowering plants and that of mammals.

Difference: _____

Similarities: _____

(3 marks)

Total 25 marks

2. If a human gets trapped in a room with limited oxygen he/she will not survive for very long.

(a) (i) Suggest ONE reason why humans need oxygen to stay alive.

(1 mark)

(ii) Write a chemical equation which summarizes the process for which oxygen is required in living cells.

(2 marks)

(iii) Name the process at (a) (ii) above.

(1 mark)

(iv) In which cell organelle does the process at (a) (iii) take place?

(1 mark)

(b) (i) During his spectacular sprint, Usain Bolt's muscle cells became starved of oxygen. Explain why this happens in exercising muscles.

(3 marks)

(ii) What substance is produced in Bolt's muscles which were starved of oxygen?

(1 mark)

(iii) Explain how the substance in (b) (ii) above could affect Bolt's performance if he continued running for another 10 minutes.

(2 marks)

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- (c) (i) Some plant species are able to survive in conditions where oxygen is lacking. Name the process by which these plants get their energy.

(1 mark)

- (ii) Describe the process named in (c) (i) above and state TWO substances produced by this process.

(3 marks)

Total 15 marks

3. Figure 1 represents a concept map of the energy flow in an ecosystem.

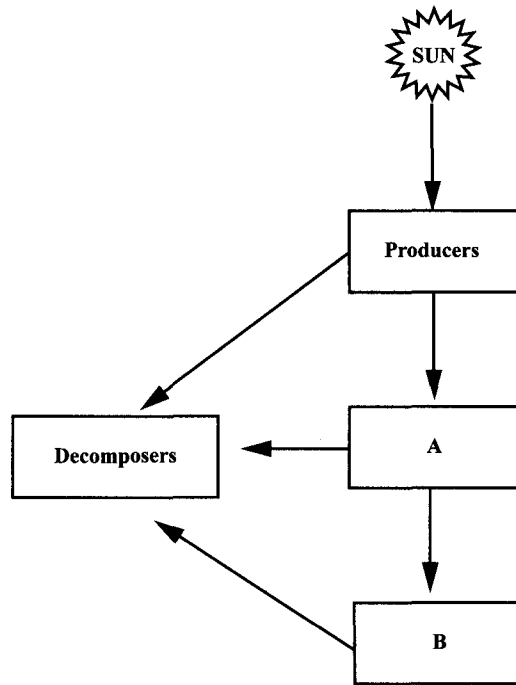


Figure 1. Concept map of energy flow in an ecosystem

- (a) Explain the unique role of producers in the ecosystem.

(1 mark)

- (b) (i) Identify A and B.

A _____
B _____
(2 marks)

- (ii) Using a named example, explain the role of B in the ecosystem.

Example: _____
Role: _____

(2 marks)

- (iii) Give ONE example of a decomposer and explain how a decomposer releases carbon dioxide to the atmosphere.

Example: _____

Carbon dioxide: _____

(3 marks)

- (iv) Suggest why energy from the sun flows in one direction while materials are cycled within the ecosystem.

(2 marks)

- (c) In addition to energy, materials are also cycled through various trophic levels and the abiotic environment. Using a diagram of the nitrogen cycle, show FOUR **named** processes which assist in recycling nitrogen in the ecosystem.

(4 marks)

- (d) State the name of the organism responsible for ONE of the processes named in (c) above.

(1 mark)

Total 15 marks

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SECTION B

Answer ALL questions. Write your answers in the spaces provided at the end of each question.

4. (a) Describe TWO ways in which a leaf is adapted for the process of photosynthesis. **(2 marks)**
- (b) Some students carried out an investigation to find out whether chlorophyll is needed for photosynthesis. They took a de-starched, potted plant with variegated (green and white) leaves and left it in sunlight for a few hours. A student then removed one leaf from the plant and tested it for starch. The student drew the leaf before the starch test.
- (i) Make a large, clearly labelled drawing to show the distribution of green and non-green areas of one leaf that the student may have used.
- (ii) Indicate, on the same drawing, the part expected to change colour if the leaf had photosynthesized.
- (iii) Describe the colour change expected if the leaf tested positive for starch. **(7 marks)**
- (c) Explain how this investigation could indicate that chlorophyll is needed for photosynthesis. Include the role of chlorophyll in your explanation. **(4 marks)**
- (d) Give an explanation for the presence of starch in the parts of a plant that do NOT contain chlorophyll. **(2 marks)**

Total 15 marks

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5. The kidney is an important excretory organ in mammals.

(a) Name TWO substances excreted by the kidney. **(2 marks)**

(b) With the aid of an annotated diagram, explain how urine is formed in the human kidney. **(9 marks)**

(c) An analysis of a sample of urine shows the presence of blood cells and glucose. Account for the presence of these substances in the urine. **(4 marks)**

Total 15 marks

Space for drawing

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6. (a) (i) Complete the following table by giving ONE example of EACH of the following categories of disease:

Category of Disease	Example of Disease
Pathogenic	
Nutritional deficiency	
Hereditary	
Physiological	

- (ii) Compare the methods used in treating and controlling the pathogenic and physiological diseases named in (a) (i). **(8 marks)**
- (b) Plants can be genetically engineered to make them resistant to diseases.
- (i) State what is meant by ‘genetic engineering’.
- (ii) Discuss the possible social, ethical and ecological implications of genetic engineering. **(7 marks)**

Total 15 marks

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